RUM CG
Agentic Browsers

2026-09-09

Agenda

- What are Agentic Browsers?
- What can they do?
- How can they be detected?
- What does it matter for RUM?

What are Agentic Browsers?

- AI browser: Adds features like summaries, writing help, chat sidebar
- Agentic browser: LLM + browser + agent, and the LLM can navigate, click, fill forms, and complete multi-step tasks on behalf of a user without the user driving each action

Human in the loop

  - Consumer: ChatGPT Atlas (retired) / Desktop, Perplexity Comet, Dia, Opera Neon, Fellou
  - Browser Features / Extensions: Chrome Auto Browse (Gemini), Microsoft Edge Browse with Copilot, Google Project Mariner (Chrome extension), Claude for Chrome, Composite, MultiOn (extension mode)

Full autonomous

  - Developer Frameworks: Browser Use, Stagehand, Skyvern, AutoGen, Eko (Fellou's framework)
  - Cloud Browser Infrastructure: Browserbase, Steel, Hyperbrowser, Anchor Browser, Scrapybara, Bright Data Agent Browser

Where do they run?

	Examples	Detection Difficulty
Local (user's machine)	Atlas, Comet, Dia, Chrome Auto Browse, Edge Copilot, Claude for Chrome, Composite, Polar, Strawberry, Opera Neon	**Hardest** — real IP, real TLS, real cookies, real fingerprint
Cloud (remote sandbox)	ChatGPT Agent Mode, Manus, Browserbase, Steel, Runner H, Claude Computer Use	**Easier** — datacenter ASN, potential TLS mismatches, framework signals

How do they view and act?

	Agents	Detection Implication
DOM/Accessibility tree	Browser Use, Stagehand, Comet (sidebar), Atlas (sidebar)	Faster, leaves DOM access traces
Screenshots (vision)	ChatGPT Agent Mode, Claude Computer Use, Runner H	Pause-act-pause timing pattern; no DOM parsing
Hybrid (both)	Manus, Claude for Chrome, Comet (agent mode), Atlas (agent mode)	Mixed signals

What can they do?

- Navigation
  - Clicking, scrolling, typing, switching tabs
- Forms
  - Contact, government, insurance, job applications
- Shopping
  - Comparing prices, browsing multiple sites, checkout
- Data Migration
  - Extract/convert information

How can they be detected?

- User-Agent
  - Most **do not** add identifiers to UA string or Client Hints (look like Chromium)
  - Some only send tokens to the mothership, e.g. `ChatGPTBrowser` to chatgpt.com
  - Google-Agent, Manus AI notable exceptions
- Web Bot Auth
  - Only ChatGPT Atlas (discontinued) + Desktop Agent mode, Google Chrome Auto Browse / Google-Agent, Anchor Browser
- Referral
  - May change the referrer to e.g. https://www.perplexity.ai
- Often have ad / gtm / RUM blockers
  - Intentionally trying to hide that traffic from observability

How can they be detected? pt 2

- JavaScript APIs
  - `navigator.webdriver` (most disable this)
  - `HeadlessChrome` in UA (most disable this)
  - CDP (`Runtime.enable`) (most disable this)
- Behavior
  - Mouse / pointer
  - Vision timing
  - Typing patterns
  - Scrolling precision
  - DOM mutations during agent mode (overlays, highlights, progress indicators)
- Inconsistencies
  - `Sec-Fetch-*`
  - Client Hints
  - Header Ordering
- Fingerprinting
  - Network / TLS / H2
  - IP / ASN

How much traffic is from Agentic Browsers?



What does it matter for RUM?

- Many of these are actively blocking ads, GTM, third-party scripts
  - RUM gets less visibility
- Do Agents (or their users) care how long it takes to load a webpage?
  - More tolerant of page load delays / web vitals than a human?
- Web Bot Auth?
  - More suitable for Crawlers, not as much for Agents?
  - No JavaScript / RUM API?

What should the RUM community do?

- Stick our heads in the sand?

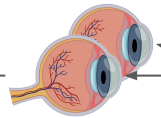
- Advocate for Agentic browsers to self-identify?
  - JavaScript API / events (could support multi-modal human<>agent handoff)
    - Best for human-in-the-loop agentic browsers where control goes back and forth
      - `canBeAgenticDriven()`
      - `isBeingAgenticDriven()`
  - User-Agent token (also available via JavaScript)
    - Best for fully-autonomous browsers
  - Client Hint header (also available via JavaScript)
    - Best for fully-autonomous browsers

Speed of User Experience

Sergey Chernyshev

Users

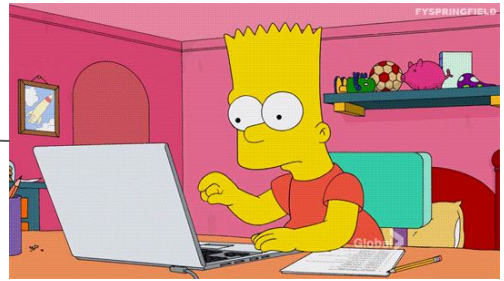




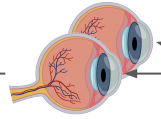




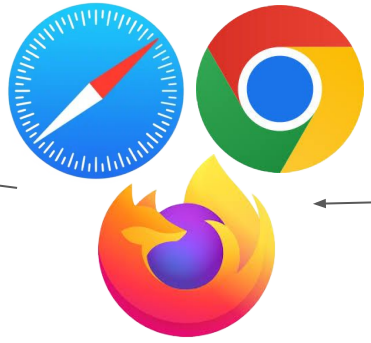
younger RUMCG members

















WEB PERFORMANCE





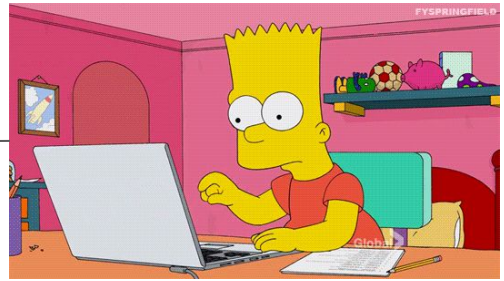
Bots







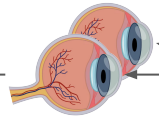





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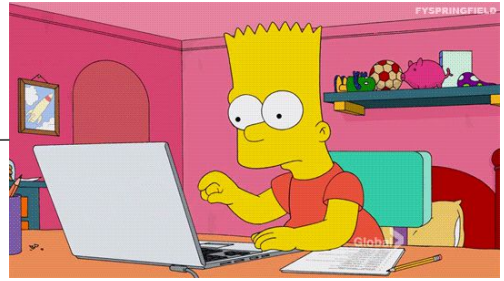




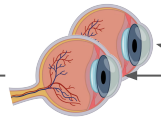


















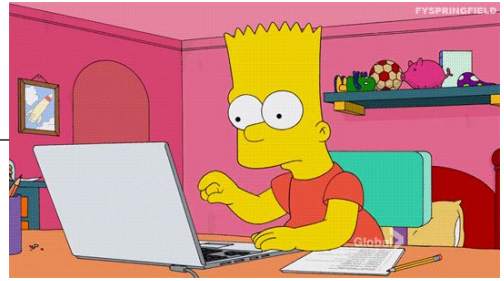

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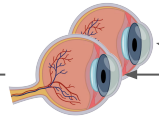





FYSPRINGFIELD

User Experience







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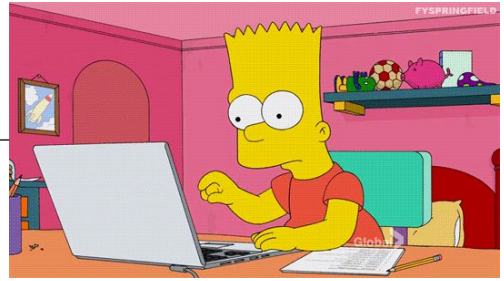




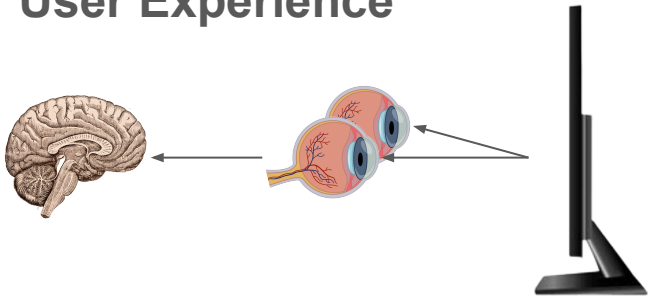








User Experience







User Experience



